

CSI ORIGIN 2026

PROBLEM STATEMENT

1 PROBLEM STATEMENT

Hackathon / Event Name: CSI ORIGIN 2026

Problem Statement ID: 1

Title: Autonomous Financial Management for Variable-Income Users

2 BACKGROUND

Current Situation / Context

Millions of individuals, particularly gig workers, freelancers, informal-sector workers, and young earners, experience highly variable income and irregular expenses. Their financial activity is fragmented across bank accounts, SMS notifications, bills, receipts, and investment records, making it difficult to maintain a consistent and accurate understanding of their overall financial position.

Traditional personal-finance applications primarily function as dashboards for tracking historical transactions. They often depend on users to manually categorise transactions, establish budgets, monitor accounts, identify recurring obligations, and interpret financial information. As a result, the user remains responsible for continuously analysing their own financial situation and deciding what action to take.

Underlying Problem or Need

The core gap is the absence of a system that can continuously construct and maintain a coherent model of an individual's evolving financial state from fragmented and heterogeneous financial information.

Such a system must account for income variability, recurring obligations, discretionary spending, upcoming expenses, savings goals, and investment activity while distinguishing between historical financial behaviour and uncertain future conditions.

Why This Is Becoming Important

Income patterns are becoming increasingly variable for gig workers, freelancers, informal-sector workers, and other non-traditional earners. At the same time, individuals are managing an increasing number of recurring payments, financial commitments, savings objectives, and investment decisions across disconnected sources.

A static financial dashboard is therefore insufficient when a user's ability to meet future obligations can change significantly with fluctuations in income or spending behaviour. Financial management increasingly requires continuous interpretation of changing circumstances rather than periodic review of historical transactions.

Limitations of the Current Situation

Existing approaches are predominantly reactive and require substantial user involvement. They may depend on manual transaction categorisation, predefined budgets, explicit user queries, and periodic account reviews.

They also struggle to reason reliably when financial data is incomplete or noisy, future income is uncertain, user behaviour changes, or multiple financial objectives compete for limited resources. Generic recommendations may fail to account for an individual's preferences, financial priorities, and risk tolerance.

Impact of Leaving This Unresolved

Without a system capable of continuously interpreting changing financial conditions, users may identify financial risks only after they materialise. They may miss upcoming obligations, underestimate the impact of variable income, continue unnecessary recurring expenses, fail to adapt savings decisions to changing circumstances, or make financial decisions without considering their broader financial state.

3 PROBLEM

What Is Going Wrong / Missing?

There is no sufficiently autonomous financial management layer capable of continuously perceiving fragmented financial information, maintaining an evolving representation of a user's financial state, reasoning about future financial conditions, and proactively determining when assistance or intervention is appropriate.

The missing capability is the transition from **financial tracking to autonomous financial decision support**. The system must be able to interpret what has happened, reason about what is likely to happen, identify emerging risks or opportunities, and determine appropriate next actions without requiring the user to explicitly initiate every interaction.

Who or What Is Affected?

The problem primarily affects individuals with variable or irregular income, including gig workers, freelancers, informal-sector workers, and young earners.

It also affects any financial management workflow that relies on fragmented financial data, including transaction records, SMS notifications, bills, receipts, savings goals, and investment records.

Why Is It Difficult or Important?

Financial decision-making requires reasoning across multiple interacting and uncertain variables. Income may fluctuate, expenses may be irregular, obligations may recur at different intervals, and user priorities may conflict.

The system must also operate despite incomplete or noisy data and changing user behaviour. Incorrect assumptions can lead to inappropriate recommendations, while excessive alerts or interventions can overwhelm users with limited attention.

Therefore, the challenge is not simply to analyse financial data, but to maintain an evolving financial model and make contextual decisions under uncertainty while remaining aligned with the user's stated preferences and risk tolerance.

Consequences

If this problem persists, financial management will remain largely reactive and dependent on continuous manual intervention. Users may fail to anticipate cash-flow shortfalls, overlook abnormal spending or recurring expenses, delay important financial decisions, or allocate available funds without adequately considering competing financial objectives.

4 CHALLENGE

Design and build an **agentic AI financial management system** that continuously transforms fragmented and heterogeneous financial information into an evolving model of a user's financial state and uses that model to proactively identify risks, opportunities, and appropriate financial actions.

The system should be capable of perceiving relevant financial changes, reasoning over current and projected conditions, prioritising competing objectives, and providing contextual and explainable decision support while operating under uncertainty and incomplete information.

Problem Requirements / Constraints

1. The system must be capable of integrating and reasoning over heterogeneous financial information, including transaction data, SMS notifications, bills, receipts, and investment records.
2. The system must account for variable income, recurring obligations, discretionary spending, upcoming expenses, savings goals, and investment activity when evaluating the user's financial state.
3. The system must identify relevant emerging financial risks or opportunities proactively rather than relying exclusively on explicit user queries or instructions.
4. Recommendations must be contextual, explainable, and aligned with the user's stated preferences and risk tolerance rather than being generic financial advice.
5. The system must remain useful when financial data is incomplete or noisy, future income is uncertain, user behaviour changes, or multiple financial objectives compete for available resources.

5 ANNEXURE — DETAILED RULES / CONSTRAINTS

Financial Data Interpretation

The system must account for the possibility that information received from different financial sources may be incomplete, inconsistent, duplicated, delayed, or noisy. It must avoid treating uncertain or inferred information as confirmed financial facts without appropriate qualification.

Income Variability

The system must not assume a fixed or predictable income pattern for variable-income users. Financial projections and recommendations should account for uncertainty in future earnings.

Financial Obligations

The system should distinguish between recurring or expected financial obligations and discretionary expenditure when reasoning about available funds and future cash flow.

Competing Objectives

The system must be capable of reasoning when multiple financial objectives compete for limited resources, such as meeting upcoming obligations, maintaining savings, achieving financial goals, and managing investment activity.

User Preferences and Risk

Recommendations must respect the user's stated financial preferences and risk tolerance. The system should provide sufficient reasoning or context for recommendations so that users can understand why an action or intervention is being suggested.

6 FINAL PROBLEM CONSTRAINTS

- The system must operate on fragmented and potentially incomplete or noisy financial information.
- The system must account for uncertainty in future income and changing user behaviour.

- The system must prioritise competing financial objectives rather than evaluating financial decisions in isolation.
- The system must provide contextual and explainable decision support aligned with user preferences and risk tolerance.
- The challenge is focused on autonomous financial decision support and proactive assistance, not merely financial transaction tracking or dashboard visualisation.